1.12 - PQ-Formel

(a) Herleitung

$$x^2 + px + q = 0$$

quadratische Ergänzung

$$\Rightarrow \quad x^2 + px = -q$$

$$\Rightarrow \quad x^2 + px + \left(\frac{p}{2}\right)^2 = -q + \left(\frac{p}{2}\right)^2$$

$$\Rightarrow \quad \left(x + \frac{p}{2}\right)^2 = \left(\frac{p}{2}\right)^2 - q$$

$$\Rightarrow \quad x + p/2 = \pm\sqrt{\left(\frac{p}{2}\right)^2 - q}$$

$$\Rightarrow \quad \boxed{x_{1/2} = -p/2 \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}}$$

$\equiv D$ "Diskriminante"

(b) Summe beider Lösungen

$$x_1 + x_2 = \left(-p/2\right) + \sqrt{D} + \left(-p/2\right) - \sqrt{D} = -p$$

(c) Produkt beider Lösungen

$$x_1 \cdot x_2 = \left(-\frac{p}{2} + \sqrt{D}\right) \cdot \left(-\frac{p}{2} - \sqrt{D}\right)$$

$$= \left(-p/2\right)^2 - \left(\sqrt{D}\right)^2 \qquad \text{"3. Binomische Formel"}$$

$$= \left(p/2\right)^2 - \left(\left(p/2\right)^2 - q\right) = q$$